**Operating Systems - Lab 7 (Feb 18)**

# Java Threads

1. Write a program to create and run two threads.
   - ✓ Create a class thread1 by implementing the Runnable interface. Thread1 prints even numbers from 1 to 10. Use *for* loop.
   - ✓ Create class thread2 by extending Thread class. thread2 prints odd numbers from 1 to 10. Use *for* loop.
   - ✓ Modify the program to include sleep of 500 milliseconds (in loop) in both threads and see the output.

2. Write a Java program that outputs prime numbers. This program should work as follows: The user will run the program and will enter a number on the command line. The program will then create a separate thread that outputs all the prime numbers less than or equal to the number entered by the user.

3. Write a Java program that calculates various statistical values for a list of numbers. This program will be passed a series of numbers on the command line and will then create five separate threads. One thread will determine the average of the numbers, the second will determine the maximum value, the third will determine the minimum value, the fourth will determine the median value, and the fifth thread will determine the sum of all numbers. For example, suppose your program is passed the integers 90 81 78 95 79 72 85. Output should be:
   - ✓ The average value is 82
   - ✓ The minimum value is 72
   - ✓ The maximum value is 95
   - ✓ The sum is 580
   - ✓ The median value is 81

4. Modify the socket based SI server (simple interest server), so that the server services each client request in a separate thread.

5. Modify the socket based SI server (simple interest server), so that the server services each client request using a thread pool (number of threads = 5).